

PROJECT REVIEW

Script-Controlled ACL – Restrict Record Access Based on Field Value

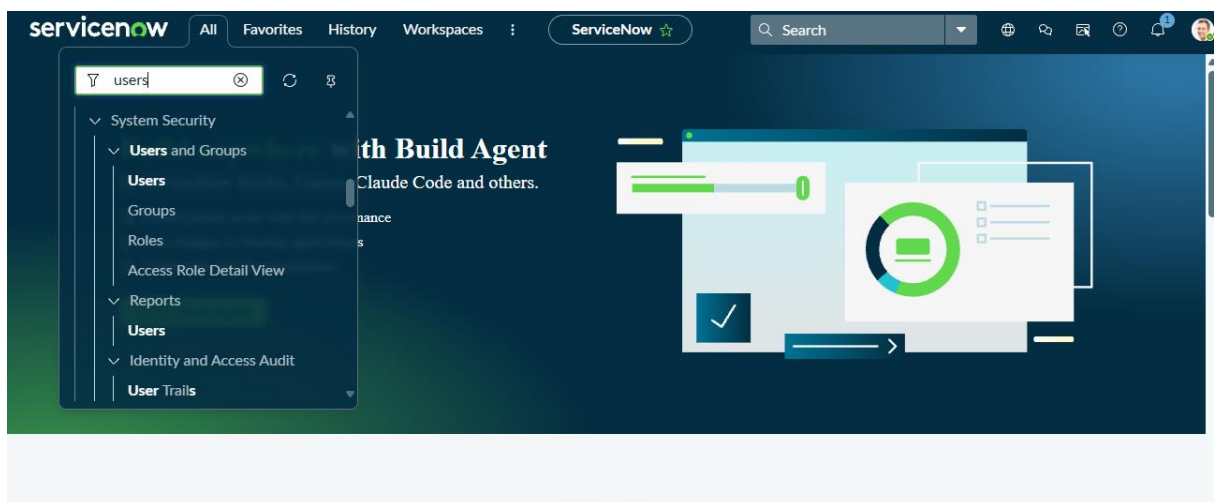
Description :

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

Milestone-1 Creation of Users and Roles

User Creation:

1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
 - Navigate to **User Administration** → **Users**
 - Click **New**
 - **User ID:** EEE User
 - **First name:** EEE
 - **Last name:** User
 - **Email:** eeeuser@gmail.com
 - Save the record



servicenow All Favorites History Admin **User - New Record** Search System Administrator: Available

User New record Submit

To set up the User's password, save the record and then click Set Password.

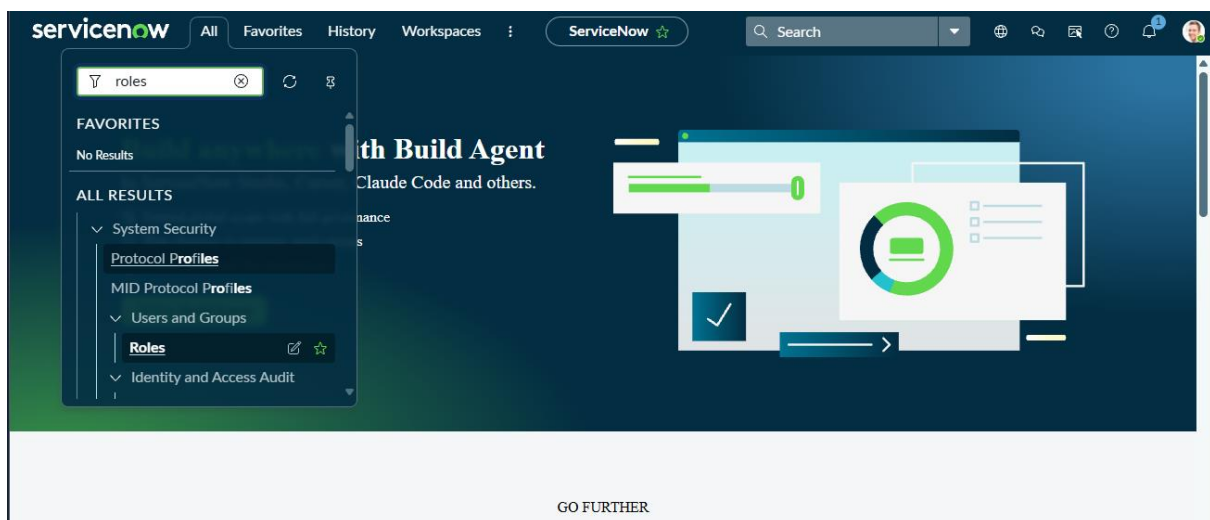
User ID	EEE User	Email	eeeuser@gmail.com
First name	EEE	Language	-- None --
Last name	User	Calendar integration	Outlook
Title		Time zone	System (America/Los_Angeles)
Department		Date format	System (2026-08-19)
Password needs reset	☐	Business phone	
Locked out	☐	Mobile phone	
Active	☒	Photo	Click to add...
Identity type	Human		
Internal Integration User	☐		

Submit

Related Links

Role Creation

- Navigate to **User Administration** → **Roles**
- Click **New**
- **Name:** Enter role name as bb1
- **Description:** Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User

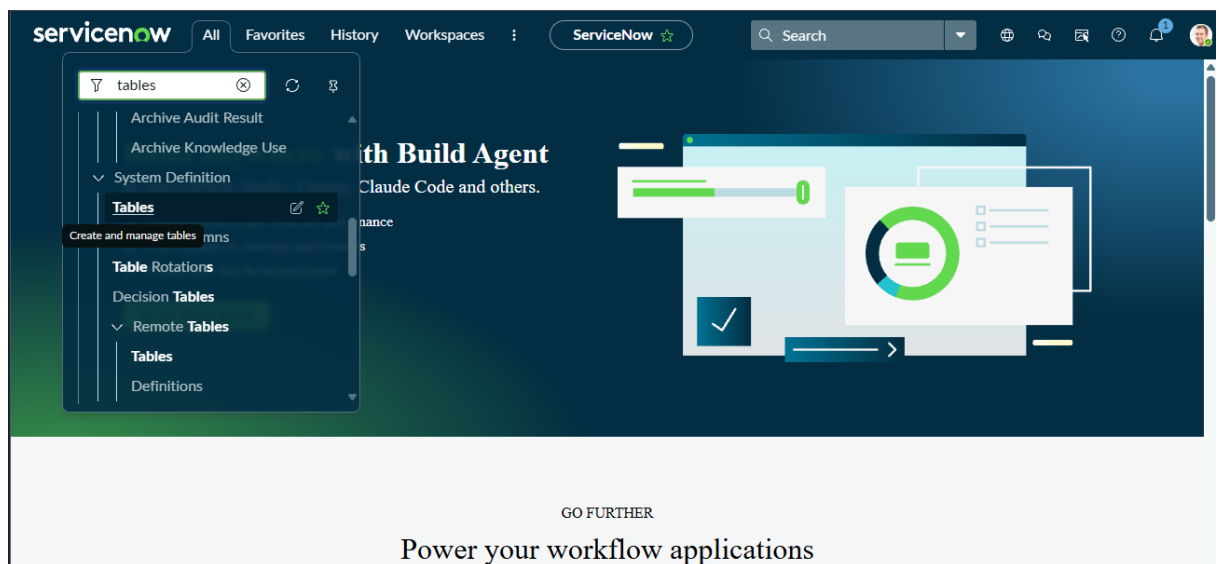


The screenshot shows the ServiceNow interface for configuring a role named 'bb1'. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The role name 'bb1' is displayed in a dropdown menu. The configuration form includes fields for 'Name' (bb1), 'Application' (Global), and 'Elevated privilege' (a checkbox). A 'Description' field is also present. At the bottom, there are 'Update' and 'Delete' buttons.

Milestone-2 Table Creation

Tables Creation

1. In the **Application Navigator**, search for **System Definition** → **Tables**.
2. Click **New** and create a custom table with the following details:
 - **Label:** Institution Details
 - **Name:** u_institution_details
 - **Extends for :** false
3. Click **Submit**.



servicenow All Favorites History Admin : Table - Institution Details ☆ Search

Table Institution Details [Delete] [Update] [Delete All Records]

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Application Global ⓘ

* Name Remote Table

Columns Controls Application Access

4. Add the following fields to the **u_institution_details** table:

- **Student Roll Number**– Auto Number
- **Student Name** – Reference- User
- **Faculty Name** – Reference- User
- **Branch** – Choice (ECE, EEE, CSE)
- **Email**– String
- **Phone Number**– String
- **Description**– Multi String

5. Save the Table

servicenow All Favorites History Admin : Table - Institution Details ☆ Search

Table Institution Details [Delete] [Update] [Delete All Records]

Table Columns for text Search 1 to 6 of 6 [New]

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
	Updated by	String	(empty)		40	false
ⓘ	Updated	Date/Time	(empty)		40	false
	Updates	Integer	(empty)		40	false
	Created by	String	(empty)		40	false
	Sys ID	Sys ID (GUID)	(empty)		32	false
	Created	Date/Time	(empty)		40	false
✕	Student Roll Number	String				false
✕	Student Name	Reference	User			false
✕	Faculty Name	Reference	User			false
✕	Branch	Choice				false
✕	Email	String				false
✕	Phone Number	String				false
✕	Description	String				false
+	Insert a new row...					

6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

servicenow All Favorites History Admin Institution Details Search							
Institution Details Branch Search Actions on selected rows... New							
All							
☐	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
	ECE			Abraham Lincoln		Abel Tutor	STU100
	CSE			Andrew Och		Amos Linnan	STU300
	EEE			Allyson Gillispie		Alejandra Prenatt	STU200

Milestone-3 Creation of Access Control List - READ

Creation of Access Control Lists(Read ACL)

1. In the **Application Navigator**, search for **System Security** → **Access Control (ACL)**.
2. Elevate “Security_admin” role
3. Click **New** to create a new ACL.
4. Enter the following details:
 - **Type:** record
 - **Operation:** read
 - **Name:** u_institution_details
 - **Active:** true
 - **Advanced:** true
5. In the **Requires role** related list, add:
 - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

```
(function () {
    // Allow admin users full access
    if (gs.hasRole('admin')) {
        return true;
    }
})
```

```
}  
// Allow only EEE branch users to see EEE records  
if (gs.hasRole('bb1')){  
    return true;  
}  
// Deny access for all others  
return false;  
})();
```

8. Click **Submit** to save the ACL.
9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name u_institution_details.list/LIST and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list.

Verify :

- User with **bb1 role** can view **only EEE branch records**
- User without the role cannot view any records
- Admin user can view **all records regardless of branch**

Search

System Administrator



System Administra...

System Administrator

ServiceNow

 Profile

 Preferences

 Keyboard shortcuts

 Impersonate user

 Elevate role

 Printer friendly version

 Log out

Student Name

Abel Tuter

Amos Linnan

Alejandra Prenatt

Abraham Lincoln

Abel Tuter

Elevate role

×

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel

Update

servicenow

AllFavoritesHistoryWorkspacesServiceNow

Search

access cont

FAVORITES

No Results

ALL RESULTS

System Security

Access Control (ACL)

Machine Identity Access Control

IP Address Access Control

with Build Agent

Claude Code and others.

GO FURTHER

Power your workflow applications

servicenow

AllFavoritesHistoryAdminAccess Controls

Search

Actions on selected rows...

New

Name	Decision Type	Operation	Type	Active	Updated by	Updated
\$allappsmgmt	Allow If	read	ui_page	true	admin	2019-02-20 01:02:07
\$atf_page_inspector	Allow If	read	ui_page	true	admin	2020-03-20 15:16:15
\$conversation-builder	Allow If	read	ui_page	true	admin	2019-06-14 15:13:38
\$mycompanyappsmgmt	Allow If	read	ui_page	true	admin	2019-02-21 02:31:44
\$products	Allow If	read	ui_page	true	admin	2020-03-10 03:19:22
\$pwd_reset_serviceDesk	Allow If	read	ui_page	true	admin	2023-05-29 05:26:27
\$sa_service_model_json	Allow If	read	ui_page	true	admin	2015-10-28 02:03:33
\$sla_timeline	Allow If	read	ui_page	true	admin	2016-11-14 01:14:39
\$sn-va-web-client-app	Allow If	read	ui_page	true	admin	2018-03-06 10:20:09
\$sn_global_search_results	Allow If	read	ui_page	true	admin	2020-06-15 10:43:57
\$spd	Allow If	read	ui_page	true	admin	2016-01-19 12:53:23
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39
\$studio	Allow If	read	ui_page	true	admin	2016-01-09 16:21:38
\$upgrade_client	Allow If	read	ui_page	true	admin	2025-11-10 16:44:04
\$upgrade_monitor	Allow If	read	ui_page	true	admin	2025-09-29 22:40:08

servicenow

AllFavoritesHistoryAccess Control - New Record

Search

Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type

record

* Operation

read

Admin overrides

Decision Type

Allow If

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

Description

Application

Global

Active

Advanced

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

Data Condition

Condition [No. of records matching the condition: 3](#)

[Add Filter Condition](#) [Add OR Clause](#)

Branch is EEE [AND](#) [OR](#) [X](#)

Advanced Condition

Advanced Condition

Controlled by References [🔒](#)

Script ☒ Turn on ECMAScript 2021 (ES12) mode ⓘ

```

1  (function () {
7      if (gs.hasRole('bb1')){
8          return true;
9      }
10     // Deny access for all others
11     return false;
12 })();
13

```

Milestone-4 Creation of Access Control List - CREATE

Creation of Access Control Lists(Create ACL)

- ❖ Click **New** to create a new ACL(Create)
- ❖ Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
- ❖ In the **Requires role** related list, add:
 - bb2 (custom role)
- ❖ No need to add any Data condition
- ❖ Save the ACL Record

Verify :

- User with **bb1 &bb2 roles** can view **only EEE branch records b New Button**

The screenshot shows the ServiceNow interface for configuring an Access Control rule. The breadcrumb trail is 'Access Control - u_institution_details'. The form fields are as follows:

- * Type: record
- * Operation: create
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: Allow create for records in u_institution_details, for users with role bb2.
- Application: Global
- Active: ☒
- Advanced: ☐

Below the form is a 'Conditions' section with a blue informational box:

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

At the bottom, there is a 'Requires role' section with a pagination indicator '1 to 1 of 1'.

Milestone-5 Creation of Access Control List - WRITE

Creation of Access Control Lists(Write ACL)

12. Click New to create a new ACL(Write)
13. Enter the following details:
 - o Type: record
 - o Operation: Write
 - o Name: u_institution_details
 - o Active: true
14. In the Requires role related list, add:
 - o bb3 (custom role)
15. No need to add any Data condition
16. Save the ACL Record

Verify :

- User with **bb1 &bb2 &bb3 roles** can view and edit the **EEE branch records along with New Button**

The screenshot shows the ServiceNow 'Access Controls' configuration page for the record 'u_institution_details'. The form includes the following fields and values:

- Type:** record
- Operation:** write
- Application:** Global
- Active:** ☒
- Advanced:** ☐
- Admin overrides:** ☒
- Decision Type:** Allow If
- Protection policy:** -- None --
- Name:** Institution Details [u_institution_details]
- Description:** Allow write for records in u_institution_details, for users with role bb3.

At the bottom of the form, there is a tab labeled 'Conditions'.

Milestone-6 Creation of Access Control List - DELETE

Creation of Access Control Lists(Delete ACL)

17. Click **New** to create a new ACL(Delete)
18. Enter the following details:
 - **Type:** record
 - **Operation:** Delete
 - **Name:** u_institution_details
 - **Active:** true
19. In the **Requires role** related list, add:
 - bb4 (custom role)
20. No need to add any Data condition
21. Save the ACL Record

Verify :

- Users with **bb1 &bb2 &bb3&bb4 roles** can see, create and edit along with delete the **EEE branch records**.

The screenshot shows the ServiceNow 'Access Control' configuration page for the record 'u_institution_details'. The configuration includes the following fields:

- * Type: record
- * Operation: delete
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: Allow delete for records in u_institution_details, for users with role bb4.
- Application: Global
- Active: ☒
- Advanced: ☐

At the bottom, there is a 'Conditions' section.

Output Image:

The screenshot shows the 'Institution Details' list view in ServiceNow. The table has the following columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. One record is displayed with the following data:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200

Milestone-7 Conclusion or Outcome

Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.